

Optimal Gram-Gauge Pairing

Ordered Generalized Eigenframes and the Tail-Inflation Barrier

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Abstract

RH-120 transfers a relative tail constant through any gauge satisfying two Loewner comparisons, but leaves the gauge choice open. We solve the exact recent-Gram problem. Let $G, G' \succ 0$, $D, D' \succ 0$, and let $\alpha_1 \leq \dots \leq \alpha_4$ and $\beta_1 \leq \dots \leq \beta_4$ be the generalized eigenvalues of (D, G) and (D', G') . Among all S satisfying $S^*GS = G'$, the least b for which $D' \preceq bS^*DS$ is

$$b_{\text{opt}} = \max_i \frac{\beta_i}{\alpha_i}.$$

It is attained by aligning the ordered generalized eigenframes. A 96-pair five-scale audit has zero theorem failures, but exposes a finite barrier: only 25 pairs have $b_{\text{opt}} < 1$, the median is 27.6936, and a weak-direction mismatch produces a conditioned maximum near 9.88×10^{290} . Thus exact recent-Gram alignment is optimally solved but does not by itself supply a uniform physical recurrence.

1 The minimax gauge problem

For a positive pair (G, D) define its relative tail matrix

$$A = G^{-1/2}DG^{-1/2}.$$

The squared RH-114 Rayleigh constant is $\lambda_{\max}(A)$. Given a target pair (G', D') , RH-120 asks for a, b, S such that $G' \succeq aS^*GS$ and $D' \preceq bS^*DS$. We first impose exact recent alignment, $S^*GS = G'$, so $a = 1$, and minimize the remaining tail factor.

Every exact-Gram gauge has the form

$$S = G^{-1/2}UG'^{1/2}, \quad U^*U = I. \tag{1}$$

Indeed, (1) gives exact alignment, and conversely $U = G^{1/2}SG'^{-1/2}$ is unitary. With $B = G'^{-1/2}D'G'^{-1/2}$, the tail comparison becomes

$$B \preceq bU^*AU. \tag{2}$$

2 Optimal ordered-eigenframe theorem

Theorem 1 (Optimal exact-Gram gauge). *Let $A, B \succ 0$ have ascending eigenvalues $\alpha_1 \leq \dots \leq \alpha_n$ and $\beta_1 \leq \dots \leq \beta_n$. Then*

$$\min_{U^*U=I} \inf\{b : B \preceq bU^*AU\} = \max_{1 \leq i \leq n} \frac{\beta_i}{\alpha_i}.$$

An optimizer maps an ordered eigenbasis of B to the correspondingly ordered eigenbasis of A .

Proof. If (2) holds, monotonicity of ordered Hermitian eigenvalues gives $\beta_i \leq b\alpha_i$ for every i ; hence every unitary has $b \geq \max_i \beta_i/\alpha_i$. Choose eigenvector matrices V_A, V_B and set $U = V_A V_B^*$. Then $U^*AU = V_B \text{diag}(\alpha_i) V_B^*$, while $B = V_B \text{diag}(\beta_i) V_B^*$. The comparison holds exactly for $b = \max_i \beta_i/\alpha_i$. $\square$

The only external ingredients are standard congruence calculus and ordered eigenvalue monotonicity [Bhatia, 2007, Horn and Johnson, 1991].

Combining Theorem 1 with RH-120 yields

$$\gamma' \leq \sqrt{b_{\text{opt}}} \gamma.$$

The induced gamma bound need not be sharp because the ratio attaining b_{opt} may occur below the top generalized mode. It is sharp when the maximal ordered ratio occurs at an index also determining the source top eigenvalue.

Corollary 2 (Semidefinite obstruction). *In the limiting semidefinite problem, if some ordered source value α_i is zero while $\beta_i > 0$, no finite exact-Gram tail factor can exist.*

This is the mechanism behind very large conditioned values: a direction that is nearly invisible to the source tail can carry non-negligible target tail energy.

3 Finite five-scale pairing audit

We recompute the RH-114 packet-block Gramians on the five archived scales. Adjacent scales are paired by side, threshold, and four normalized phases, giving $4 \times 2 \times 3 \times 4 = 96$ pairs. To make the generalized pencils numerically definite, both recent and tail Gramians receive an explicitly recorded relative 10^{-12} spectral floor after outward PSD correction. This defines a regularized finite diagnostic; it is not silently treated as a lower support certificate.

For every pair we construct the ordered-eigenframe gauge, verify exact Gram alignment, verify the Loewner tail upper, compare both Rayleigh constants, and enumerate all 24 eigenvalue permutations. There are zero failures in all four categories. The scale-edge results are

Across all pairs, 25 are contractive, the median is 27.693586, and every target regularized gamma remains below one. The optimal transferred upper is often loose because a lower generalized direction, rather than the top one, controls b_{opt} .

edge	pairs with $b_{\text{opt}} < 1$	median b_{opt}	maximum b_{opt}
$0.16 \rightarrow 0.08$	4/24	3.3753×10^4	9.88×10^{290}
$0.08 \rightarrow 0.04$	3/24	1.0863×10^5	3.69×10^8
$0.04 \rightarrow 0.02$	15/24	0.881856	230.059
$0.02 \rightarrow 0.01$	3/24	27.4686	4.2721×10^4

Table 1: Finite optimal tail inflation. The enormous first-edge maximum is a conditioned weak-mode diagnostic, not an asymptotic claim.

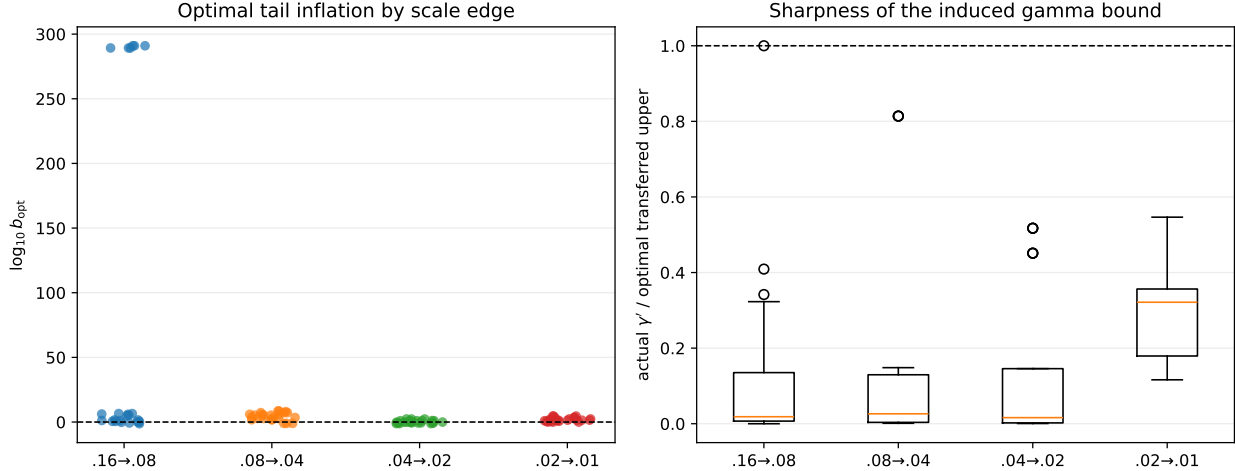


Figure 1: Left: intrinsic exact-Gram tail inflation. Right: efficiency of the resulting gamma upper.

4 What is closed and what remains

The gauge-choice problem is closed at the exact finite-dimensional level: no different exact alignment can beat the ordered generalized-spectrum formula. The audit also gives a useful negative answer to a tempting next step. Recent-Gram alignment alone does not produce a small or monotone tail factor on the archived chain. A physical recurrence must control the whole relative pair, permit explicit defects, or exploit additional dynamics.

RH-122 isolates the still sharper fixed-coordinate obstruction. RH-123 then allows additive Loewner defects, which is the form needed for outward transport. No finite extrema above are extrapolated to all scales. No uniform Stage A, Hilbert–Polya operator, zeta-zero identification, or Riemann Hypothesis conclusion is claimed.

References

- Rajendra Bhatia. *Positive Definite Matrices*. Princeton University Press, 2007.
- Roger A. Horn and Charles R. Johnson. *Topics in Matrix Analysis*. Cambridge University Press, 1991.